SAT 11

Math Module 1

Question #	ID			Answer
SAT11 M 1.1	a11ma101	$$4x+1=33$$ Which equation has the same solution as the given equation?	A. $4x=32$ B. $4x=5$ C. $4x=1$ D. $4x=-32$	A
SAT11 M 1.2	a11ma102	The function k is defined by $k(x)=x^3+110$. What is the value of $k(x)$ when $x=2$?	A. 118 B. 116 C. 115 D. 110	A
SAT11 M 1.3	a11ma103	(table below)		8

Task	Time (minutes)
I	5
II	2
III	15
IV	9
V	9

The table shows the amount of time it took Robert to complete each of the 5 tasks in a project. What was the mean time, in minutes, for Robert to complete a task for this project?

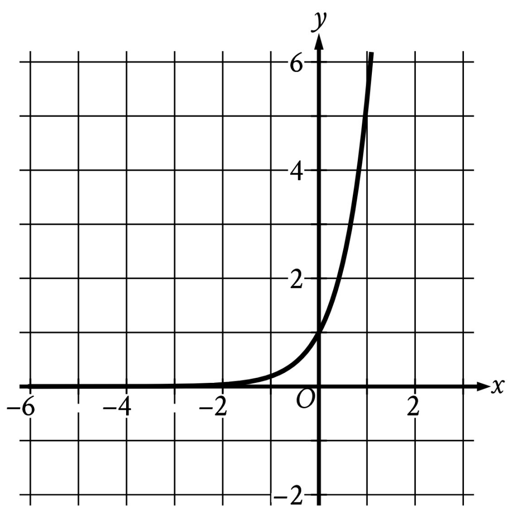
Question #	ID		Answer
SAT11 M 1.4	a11ma104		13

$$8x + 11y = 170$$

The equation gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x, and the number of 1996 select grade Lincoln Pennies, y, in a collection that is worth a total of $\$170$. If there are 6 1996 select grade Lincoln Pennies in the collection, how many 2009 premium grade Log Cabin Pennies are in the collection?

SAT11 M 1.5 a11ma105 — Answer: B

The graph of the function f is shown, where $y = f(x)$.



Which of the following best describes the function f?

A. Decreasing exponential
B. Increasing exponential
C. Decreasing linear
D. Increasing linear

SAT11 M 1.6 a11ma106 — Answer: C

Which expression is equivalent to $6x + 5x + 4y$?

A. $15x$
B. $15y$
C. $11x + 4y$
D. $30x + 4y$

Question #	ID		Answer
SAT11 M 1.7	a11ma107	Triangles ABC and DEF are congruent, where A corresponds to D, and B and E are right angles. The measure of angle A is 27°. What is the measure, in degrees, of angle F?	63
SAT11 M 1.8	a11ma108	There are 240 players in a tennis competition that includes 4 rounds of matches. Each player in the competition will play a match against another player in round 1. At the end of each round, the player who loses the match is eliminated and the player who won the match advances to the next round to play a match against another winning player. Which equation gives the number of players, p, eliminated at the end of round r, where $r \leq 4$? A. $p = 15\left(\frac{1}{2}\right)^r$ B. $p = 15(2)^r$ C. $p = 240\left(\frac{1}{2}\right)^r$ D. $p = 240(2)^r$	C
SAT11 M 1.9	a11ma109	$\frac{r}{38} = s + t$ The given equation relates the quantities r, s, and t. Which equation correctly expresses r in terms of s and t? A. $r = s + t + 38$ B. $r = s + t - 38$ C. $r = 38(s + t)$ D. $r = \frac{s+t}{38}$	C
SAT11 M 1.10	a11ma110	Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy-plane and passes through the point $(0, 5)$. Which equation defines line j? A. $y = 6x + 5$ B. $y = -5x + 5$ C. $y = -6x + 5$ D. $y = 5x + 5$	A

Question #	ID			Answer
SAT11 M 1.11	a11ma111	The length of one side of square M is 6 times the length of one side of square N. The area of square N is 49 square feet. What is the area, in square feet, of square M?	A. 1,764 B. 504 C. 294 D. 252	A
SAT11 M 1.12	a11ma112	Which quadratic equation has exactly one distinct real solution?	A. $(x+15)^2=0$ B. $(x+15)^2=-45$ C. $(x+15)^2=45$ D. $(x+15)^2=135$	A
SAT11 M 1.13	a11ma113	The cost to rent a bus from Company X is \$950 for the first 3 hours and an additional \$50 per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t>3$, is \$1,150, which equation represents this situation?	A. $950(t-3)+50t=1{,}150$ B. $950(3t)+50t=1{,}150$ C. $950+50(t-3)=1{,}150$ D. $950+50(3t)=1{,}150$	C

Question #	ID		Answer
SAT11 M 1.14	a11ma114		D

x	$g(x)$
1	32
2	28
3	24
4	20

For the linear function g, the table shows four values of x and their corresponding values of $g(x)$. The function can be written as $g(x) = mx + b$, where m and b are constants. What is the value of b?

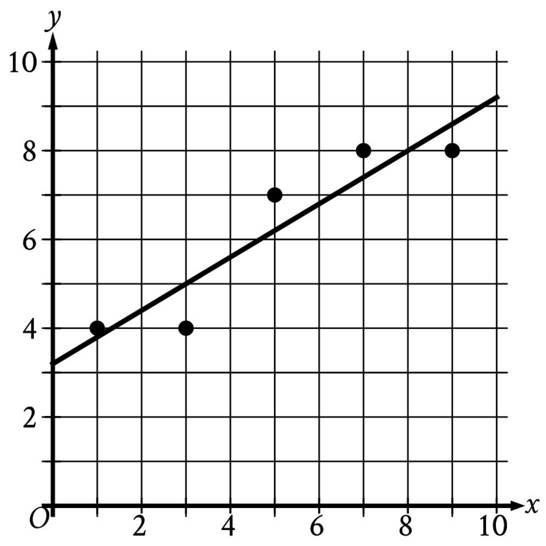
A. 4
B. 16
C. 32
D. 36

SAT11 M 1.15	a11ma115	15000

The population of the town of Smithville doubled every 75 years from 1659 to 1959. The population of this town was 240,000 in 1959. What was the population of this town in 1659?

Question #	ID			Answer
SAT11 M 1.16	a11ma116			A

The scatterplot shows the relationship between x and y. A line of best fit is also shown.



Which of the following is closest to the slope of this line of best fit?

A. 0.60
B. 2.50
C. 7.80
D. 8.00

SAT11 M 1.17 a11ma117 **B**

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f, the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f?

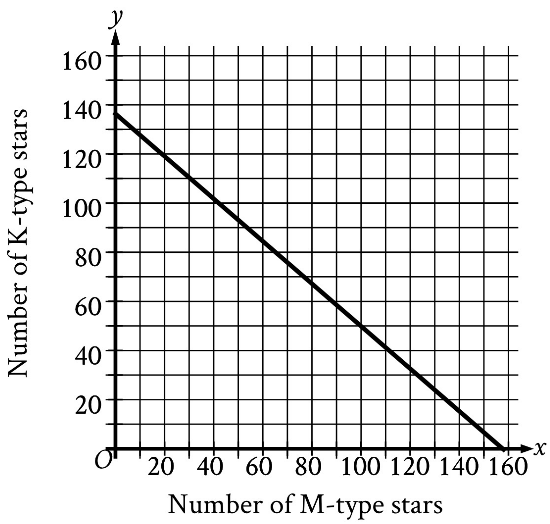
A. $f(x)=(x-24)(x-30)+4$
B. $f(x)=(x-24)(x-30)-8$
C. $f(x)=(x-8)(x-32)+32$
D. $f(x)=(x-8)(x-32)-32$

SAT11 M 1.18 a11ma118 **B**

A circle in the xy-plane has its center at $(19, 18)$ and has a radius of $9k$. Which equation represents this circle?

A. $(x-19)^2+(y-18)^2=81k$
B. $(x-19)^2+(y-18)^2=81k^2$
C. $(x-19)^2+(y-18)^2=9k$
D. $(x-19)^2+(y-18)^2=9k^2$

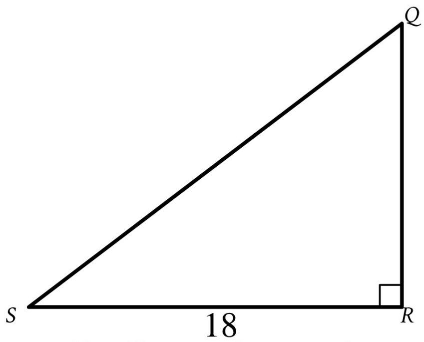
Question #	ID		Answer
SAT11 M 1.19	a11ma119		A



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is $127{,}882 \text{ quettagrams}$. The graph shown models the possible combinations of the number of M-type stars, x, and K-type stars, y, that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

A. 811
B. 938
C. 51,904
D. 75,978

SAT11 M 1.20	a11ma120		D



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

A. $18 \cos Q$
B. $18 \sin Q$
C. $\frac{18}{\cos Q}$
D. $\frac{18}{\sin Q}$

Question #	ID		Answer
SAT11 M 1.21	a11ma121	For a particular car, the linear function f gives the predicted power, in brake horsepower (bhp), for engine speeds between 1,000 revolutions per minute (rpm) and 6,000 rpm. According to this function, the car's predicted power is 433 bhp at an engine speed of 3,331 rpm and 600 bhp at an engine speed of 4,500 rpm. The equation $f(x) = \frac{1}{7}(x - a) + 433$ defines f, where x is the engine speed, in rpm, and a is a constant. What is the value of a?	3331
SAT11 M 1.22	a11ma122	$$52(x^3 + 64)(x^4 - 81) = 0$$ How many distinct real solutions does the given equation have? A. Exactly two B. Exactly three C. Exactly five D. Exactly seven	B